

PRACTICAL 6: LAN CONSTRUCTION AND SWITCH CONFIGURATION

Objective

Build a LAN using a Cisco Catalyst switch.

Equipment

- 10 PCs
- Cisco Catalyst 3560 Switch
- Patch Panel
- Fly Leads

Activities

1. Connect all 10 PCs.

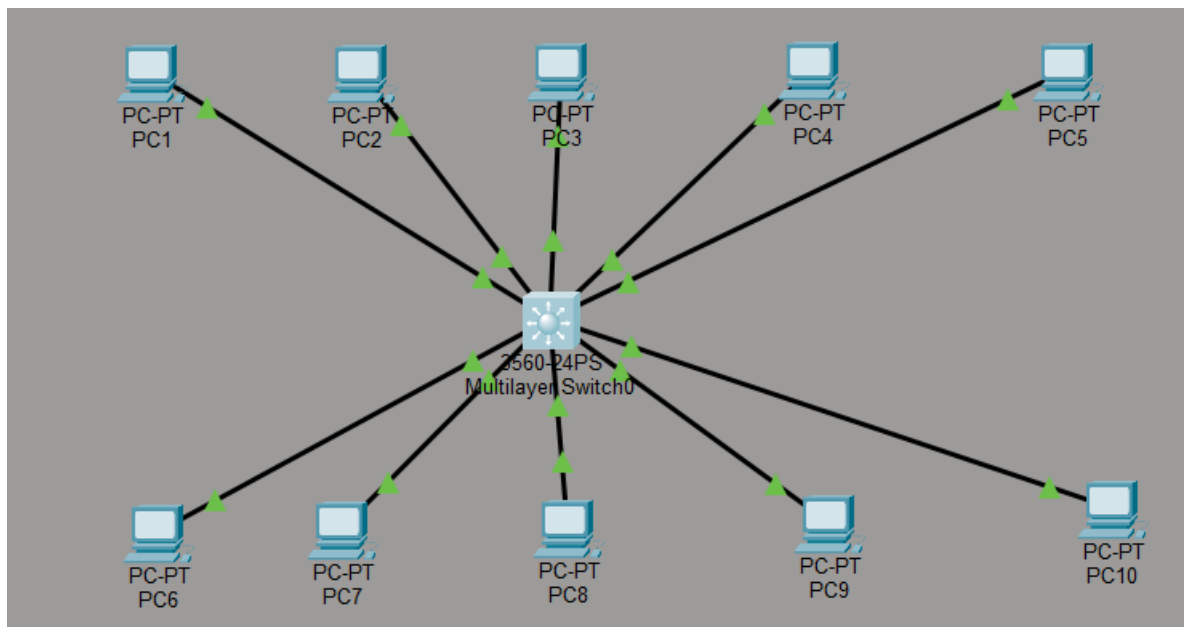


Figure 1: 10 pc's connected

2. Configure static IP addresses.

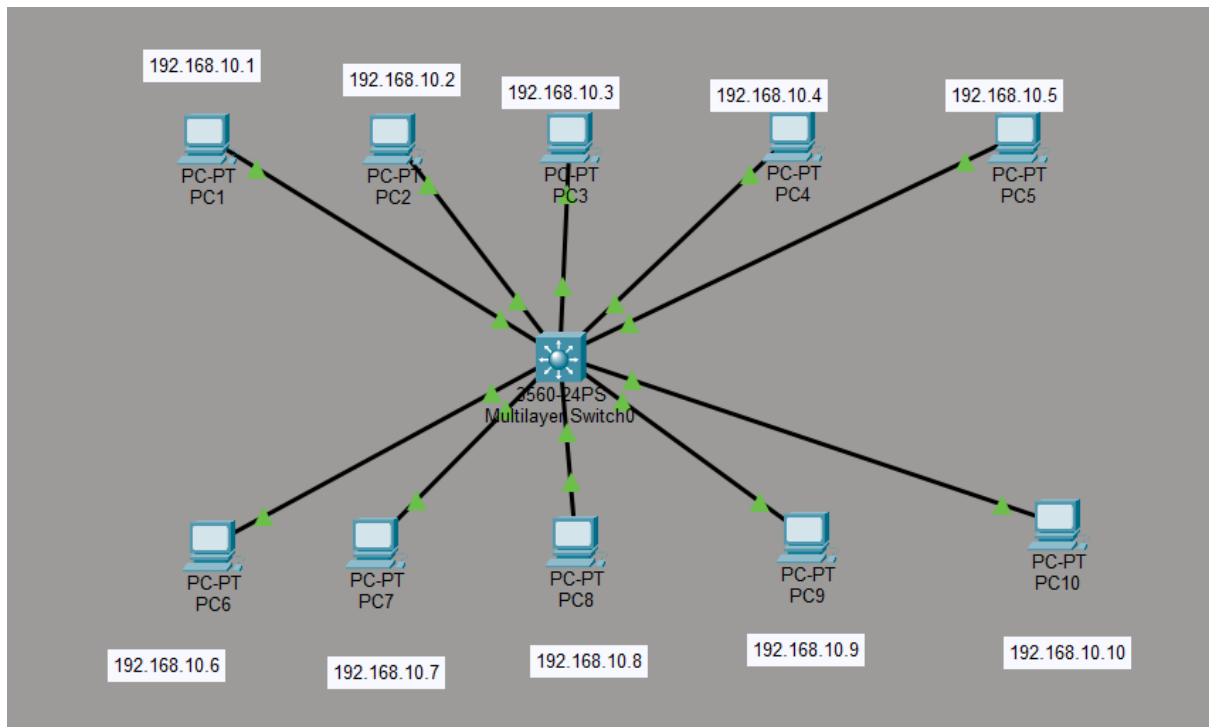


Figure 2: Static IP adressess configured

3. Configure switch management interface.

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface vlan 1
Switch(config-if)#ip address 192.168.10.254 255.255.255.0
Switch(config-if)#no shutdown

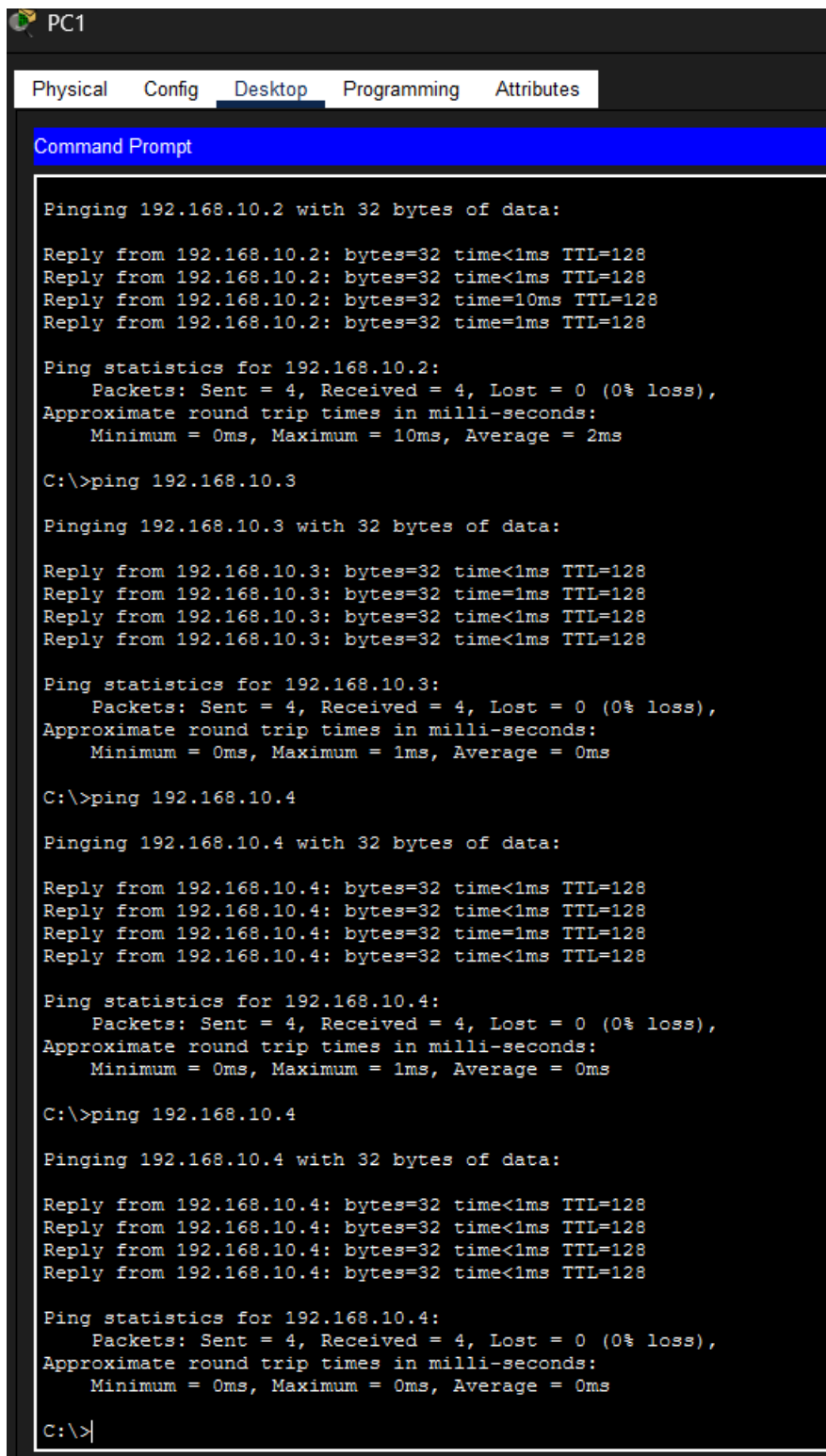
Switch(config-if)#
%LINK-3-UPDOWN: Interface Vlan1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state to up

Switch(config-if)#ip default-gateway 192.168.10.1
Switch(config)#
```

Figure 3: Switch config

4. Verify connectivity.



The screenshot shows a Windows Command Prompt window titled "PC1". The window has a menu bar with "Physical", "Config", "Desktop", "Programming", and "Attributes". The "Desktop" tab is selected. The Command Prompt displays the results of a series of ping commands. The first command is "ping 192.168.10.2", which shows four successful replies with 32 bytes of data, times less than 1ms, and a TTL of 128. The statistics for 192.168.10.2 show 4 packets sent, 4 received, 0% loss, and round trip times of 0ms, 10ms, and 2ms. The second command is "ping 192.168.10.3", which also shows four successful replies with 32 bytes of data, times less than 1ms, and a TTL of 128. The statistics for 192.168.10.3 show 4 packets sent, 4 received, 0% loss, and round trip times of 0ms, 1ms, and 0ms. The third command is "ping 192.168.10.4", which shows four successful replies with 32 bytes of data, times less than 1ms, and a TTL of 128. The statistics for 192.168.10.4 show 4 packets sent, 4 received, 0% loss, and round trip times of 0ms, 1ms, and 0ms. The fourth command is "ping 192.168.10.4" again, which shows four successful replies with 32 bytes of data, times less than 1ms, and a TTL of 128. The statistics for 192.168.10.4 show 4 packets sent, 4 received, 0% loss, and round trip times of 0ms, 0ms, and 0ms. The prompt "C:\>" is visible at the bottom of the window.

```
PC1
Physical Config Desktop Programming Attributes
Command Prompt

Pinging 192.168.10.2 with 32 bytes of data:

Reply from 192.168.10.2: bytes=32 time<1ms TTL=128
Reply from 192.168.10.2: bytes=32 time<1ms TTL=128
Reply from 192.168.10.2: bytes=32 time=10ms TTL=128
Reply from 192.168.10.2: bytes=32 time=1ms TTL=128

Ping statistics for 192.168.10.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 10ms, Average = 2ms

C:\>ping 192.168.10.3

Pinging 192.168.10.3 with 32 bytes of data:

Reply from 192.168.10.3: bytes=32 time<1ms TTL=128
Reply from 192.168.10.3: bytes=32 time=1ms TTL=128
Reply from 192.168.10.3: bytes=32 time<1ms TTL=128
Reply from 192.168.10.3: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 192.168.10.4

Pinging 192.168.10.4 with 32 bytes of data:

Reply from 192.168.10.4: bytes=32 time<1ms TTL=128
Reply from 192.168.10.4: bytes=32 time<1ms TTL=128
Reply from 192.168.10.4: bytes=32 time=1ms TTL=128
Reply from 192.168.10.4: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 192.168.10.4

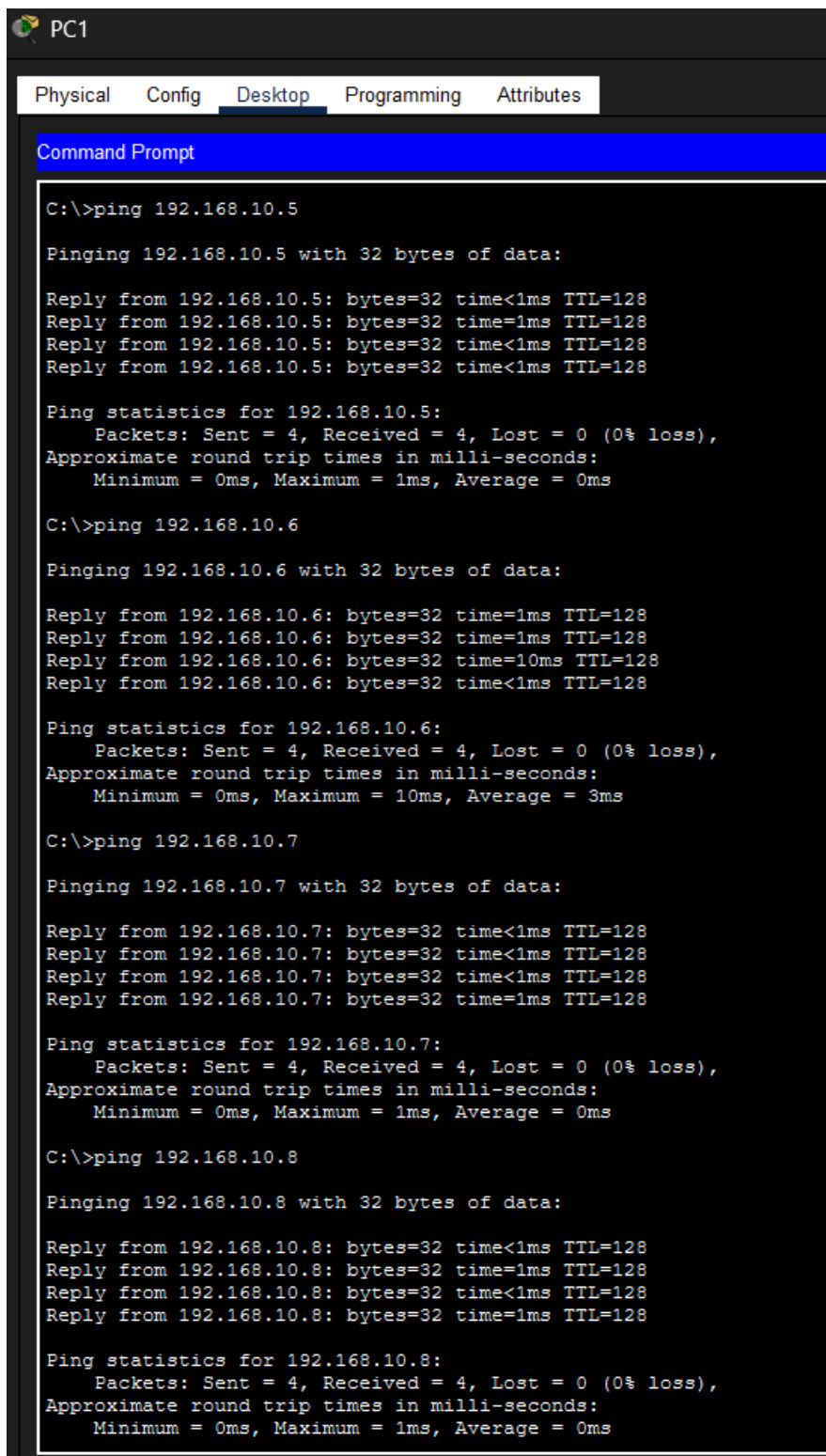
Pinging 192.168.10.4 with 32 bytes of data:

Reply from 192.168.10.4: bytes=32 time<1ms TTL=128
Reply from 192.168.10.4: bytes=32 time<1ms TTL=128
Reply from 192.168.10.4: bytes=32 time<1ms TTL=128
Reply from 192.168.10.4: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>|
```

Figure 6: ping from PC1 to PC2-4



The image shows a screenshot of a PC1 window with a Command Prompt open. The window has tabs for Physical, Config, Desktop, Programming, and Attributes, with 'Desktop' selected. The Command Prompt displays the results of four ping commands. Each command is followed by four replies and a summary of ping statistics. The statistics for all four IP addresses show 0% loss, 4 packets sent and received, and an average round trip time of 0ms.

```
C:\>ping 192.168.10.5

Pinging 192.168.10.5 with 32 bytes of data:

Reply from 192.168.10.5: bytes=32 time<1ms TTL=128
Reply from 192.168.10.5: bytes=32 time=1ms TTL=128
Reply from 192.168.10.5: bytes=32 time<1ms TTL=128
Reply from 192.168.10.5: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.5:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 192.168.10.6

Pinging 192.168.10.6 with 32 bytes of data:

Reply from 192.168.10.6: bytes=32 time=1ms TTL=128
Reply from 192.168.10.6: bytes=32 time=1ms TTL=128
Reply from 192.168.10.6: bytes=32 time=10ms TTL=128
Reply from 192.168.10.6: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.6:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 10ms, Average = 3ms

C:\>ping 192.168.10.7

Pinging 192.168.10.7 with 32 bytes of data:

Reply from 192.168.10.7: bytes=32 time<1ms TTL=128
Reply from 192.168.10.7: bytes=32 time<1ms TTL=128
Reply from 192.168.10.7: bytes=32 time<1ms TTL=128
Reply from 192.168.10.7: bytes=32 time=1ms TTL=128

Ping statistics for 192.168.10.7:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 192.168.10.8

Pinging 192.168.10.8 with 32 bytes of data:

Reply from 192.168.10.8: bytes=32 time<1ms TTL=128
Reply from 192.168.10.8: bytes=32 time=1ms TTL=128
Reply from 192.168.10.8: bytes=32 time<1ms TTL=128
Reply from 192.168.10.8: bytes=32 time=1ms TTL=128

Ping statistics for 192.168.10.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

Figure 7: ping from PC1 to PC5-8

```

C:\>ping 192.168.10.9

Pinging 192.168.10.9 with 32 bytes of data:

Reply from 192.168.10.9: bytes=32 time=1ms TTL=128
Reply from 192.168.10.9: bytes=32 time<1ms TTL=128
Reply from 192.168.10.9: bytes=32 time<1ms TTL=128
Reply from 192.168.10.9: bytes=32 time=1ms TTL=128

Ping statistics for 192.168.10.9:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 192.168.10.10

Pinging 192.168.10.10 with 32 bytes of data:

Reply from 192.168.10.10: bytes=32 time<1ms TTL=128
Reply from 192.168.10.10: bytes=32 time<1ms TTL=128
Reply from 192.168.10.10: bytes=32 time<1ms TTL=128
Reply from 192.168.10.10: bytes=32 time<1ms TTL=128

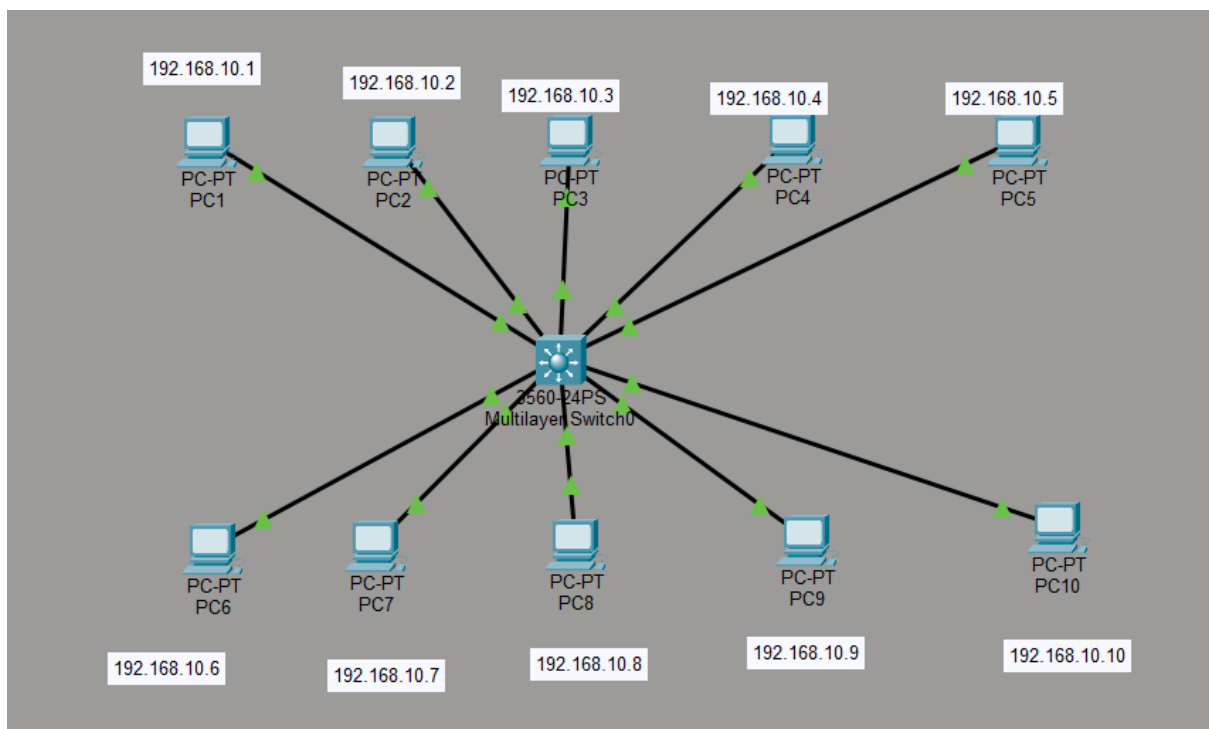
Ping statistics for 192.168.10.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>|

```

Figure 8: ping from PC1 to PC9-10

5. Create a network diagram.



IP Addressing Scheme	IP Address
Device	
PC1	192.168.10.1

PC2	192.168.10.2
PC3	192.168.10.3
PC4	192.168.10.4
PC5	192.168.10.5
PC6	192.168.10.6
PC7	192.168.10.7
PC8	192.168.10.8
PC9	192.168.10.9
PC10	192.168.10.10
Switch VLAN 1	192.168.10.25 4